

```
!pip install spacy transformers torch pandas sequeval
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Collecting sequeval
  Downloading sequeval-1.2.2.tar.gz (43 kB)
    43.6/43.6 kB 1.7 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
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Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.3)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.12.3)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.24.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.4.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: scikit-learn>=0.21.3 in /usr/local/lib/python3.12/dist-packages (from sequeval) (1.6.1)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: shellingham in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4->spacy) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4->spacy) (2.41.4)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4->spacy) (0.4.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2026.1.4)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.21.3->sequeval) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.21.3->sequeval) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.21.3->sequeval) (3.6.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (1.3.3)
```

```
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (0.1.5)
```

```
# spaCy library for NLP processing and NER
import spacy

# displacy for visualization of entities
from spacy import displacy

# Transformers for pretrained NER models
from transformers import pipeline, AutoTokenizer, AutoModelForTokenClassification

# torch for deep learning backend used by transformers
import torch

# pandas for tabular data storage
import pandas as pd
```

```
# Load English pretrained model
nlp = spacy.load("en_core_web_sm")

print("spaCy model loaded successfully")
```

```
spaCy model loaded successfully
```

```
sentences = [
    "India won the cricket match against Australia in Mumbai.",
    "Narendra Modi met Joe Biden in Washington.",
    "Microsoft announced a new AI model in 2025.",
    "Elon Musk is the CEO of Tesla and SpaceX.",
    "The Olympics will be held in Paris."
]
```

```
spacy_results = []

for sent in sentences:
    doc = nlp(sent)

    print("\nSentence:", sent)
    print("Entities:")

    for ent in doc.ents:
        print(ent.text, "->", ent.label_)

    spacy_results.append([sent, ent.text, ent.label_])
```

```
Sentence: India won the cricket match against Australia in Mumbai.
Entities:
India -> GPE
Australia -> GPE
Mumbai -> GPE
```

```
Sentence: Narendra Modi met Joe Biden in Washington.
Entities:
```

```

Narendra Modi -> PERSON
Joe Biden -> PERSON
Washington -> GPE

Sentence: Microsoft announced a new AI model in 2025.
Entities:
Microsoft -> ORG
AI -> GPE
2025 -> DATE

Sentence: Elon Musk is the CEO of Tesla and SpaceX.
Entities:
Elon Musk -> PERSON
Tesla -> ORG

Sentence: The Olympics will be held in Paris.
Entities:
Olympics -> EVENT
Paris -> GPE

```

```
df_spacy = pd.DataFrame(spacy_results, columns=["Sentence", "Entity", "Label"])
```

```
df_spacy
```

	Sentence	Entity	Label
0	India won the cricket match against Australia ...	India	GPE
1	India won the cricket match against Australia ...	Australia	GPE
2	India won the cricket match against Australia ...	Mumbai	GPE
3	Narendra Modi met Joe Biden in Washington.	Narendra Modi	PERSON
4	Narendra Modi met Joe Biden in Washington.	Joe Biden	PERSON
5	Narendra Modi met Joe Biden in Washington.	Washington	GPE
6	Microsoft announced a new AI model in 2025.	Microsoft	ORG
7	Microsoft announced a new AI model in 2025.	AI	GPE
8	Microsoft announced a new AI model in 2025.	2025	DATE
9	Elon Musk is the CEO of Tesla and SpaceX.	Elon Musk	PERSON
10	Elon Musk is the CEO of Tesla and SpaceX.	Tesla	ORG
11	The Olympics will be held in Paris.	Olympics	EVENT
12	The Olympics will be held in Paris.	Paris	GPE

```

doc = nlp(sentences[1])
displacy.render(doc, style="ent", jupyter=True)

```

```
Narendra Modi PERSON met Joe Biden PERSON in Washington GPE .
```

```
model_name = "dbmdz/bert-large-cased-finetuned-conll03-english"
```

```
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForTokenClassification.from_pretrained(model_name)

ner_pipeline = pipeline(
    "ner",
    model=model,
    tokenizer=tokenizer,
    aggregation_strategy="simple"
)

print("Transformer model loaded")
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.

To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in your Google Colab and restart the notebook. You will be able to reuse this secret in all of your notebooks.

Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
```

```
config.json: 100%                               998/998 [00:00<00:00, 21.5kB/s]
```

```
tokenizer_config.json: 100%                     60.0/60.0 [00:00<00:00, 780B/s]
```

```
vocab.txt: 213k/? [00:00<00:00, 1.60MB/s]
```

```
model.safetensors: 100%                       1.33G/1.33G [00:31<00:00, 42.4MB/s]
```

```
Loading weights: 100%                         391/391 [00:01<00:00, 266.84it/s, Materializing param=classifier.weight]
```

```
BertForTokenClassification LOAD REPORT from: dbmdz/bert-large-cased-finetuned-conll103-english
```

Key	Status	
bert.pooler.dense.weight	UNEXPECTED	
bert.pooler.dense.bias	UNEXPECTED	

Notes:

- UNEXPECTED : can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Transformer model loaded

```
transformer_results = []

for sent in sentences:
    print("\nSentence:", sent)

    output = ner_pipeline(sent)

    for entity in output:
        word = entity['word']
        label = entity['entity_group']
        score = entity['score']

        print(word, "->", label, "| Confidence:", round(score,3))

    transformer_results.append([sent, word, label, score])
```

```
Sentence: India won the cricket match against Australia in Mumbai.
India -> LOC | Confidence: 1.0
```

```
Australia -> LOC | Confidence: 1.0
Mumbai -> LOC | Confidence: 0.999

Sentence: Narendra Modi met Joe Biden in Washington.
Narendra Modi -> PER | Confidence: 0.994
Joe Biden -> PER | Confidence: 0.997
Washington -> LOC | Confidence: 0.999

Sentence: Microsoft announced a new AI model in 2025.
Microsoft -> ORG | Confidence: 0.999

Sentence: Elon Musk is the CEO of Tesla and SpaceX.
Elon Musk -> PER | Confidence: 0.999
Tesla -> ORG | Confidence: 0.997
SpaceX -> ORG | Confidence: 0.999

Sentence: The Olympics will be held in Paris.
Olympics -> MISC | Confidence: 0.992
Paris -> LOC | Confidence: 0.999
```

```
clean_results = []

for sent, word, label, score in transformer_results:

    # remove subword symbols
    word = word.replace("##", "")

    clean_results.append([sent, word, label, score])

df_transformer = pd.DataFrame(
    clean_results,
    columns=["Sentence", "Entity", "Label", "Confidence"]
)

df_transformer
```

	Sentence	Entity	Label	Confidence
0	India won the cricket match against Australia ...	India	LOC	0.999752
1	India won the cricket match against Australia ...	Australia	LOC	0.999790
2	India won the cricket match against Australia ...	Mumbai	LOC	0.999375
3	Narendra Modi met Joe Biden in Washington.	Narendra Modi	PER	0.993933
4	Narendra Modi met Joe Biden in Washington.	Joe Biden	PER	0.996672
5	Narendra Modi met Joe Biden in Washington.	Washington	LOC	0.999126
6	Microsoft announced a new AI model in 2025.	Microsoft	ORG	0.999218
7	Elon Musk is the CEO of Tesla and SpaceX.	Elon Musk	PER	0.999065
8	Elon Musk is the CEO of Tesla and SpaceX.	Tesla	ORG	0.996662
9	Elon Musk is the CEO of Tesla and SpaceX.	SpaceX	ORG	0.998991
10	The Olympics will be held in Paris.	Olympics	MISC	0.992067
11	The Olympics will be held in Paris.	Paris	LOC	0.999175

```
comparison = pd.DataFrame({
    "Feature": [
        "Model Type",
        "Speed",
        "Accuracy",
        "Context Handling",
        "Confidence Score"
    ],
    "spaCy": [
        "Statistical",
        "Fast",
        "Medium",
        "Moderate",
        "No"
    ],
    "HuggingFace": [
        "Transformer",
        "Slower",
        "High",
        "Excellent",
        "Yes"
    ]
})

comparison
```

Feature	spaCy	HuggingFace
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